Within the flat 6D Conformal Spacetime Algebra system ($Cl(4,1,1)$), an amplitude modulation (AM) detector constructed from a safety pin, graphite pencil lead, and a blue razor blade (classically known as a *foxhole radio* detector) can be modeled as a point-contact Schottky diode.

This model describes how the mechanical junction rectifies ballistic wavefront trains [5.5, 5.8] and extracts the low-frequency envelope using localized soliton transitions and passive filtering.

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**Section 5.9: The Foxhole Radio Detector (Point-Contact Oxide Rectifier)**

This detector consists of a spring-loaded metal safety pin holding a graphite (semimetal) tip in light point-contact with a "blue" steel razor blade. The "blueing" process creates a thin, chemically grown layer of iron oxide (magnetite, $Fe_3O_4$), which functions as an n-type semiconductor.

```
Safety Pin (Spring Tension)
     |
     v     /|
    ( )   / | Graphite (Pencil Lead)
     |   /  |
     |  /   v
     | /   (•) Point-Contact Junction (d_junction ~ delta)
     |/     -------------------------
     +---->|  Blued Razor Blade      |  (Iron Oxide Semiconductor)
           |_________________________|
```

**5.9.1 The Point-Contact Potential Barrier** Because the contact area between the graphite tip and the thin iron oxide layer is extremely small, it forms a microscopic junction. The depletion layer width $d_{\text{junction}}$ is comparable to the contact's close-range mechanical softening parameter [Section 3]:

$$d_{\text{junction}} \sim \delta_{\text{contact}}$$

The work function difference between the graphite semimetal and the $Fe_3O_4$ oxide semiconductor creates an asymmetric Schottky potential barrier $V_{\text{barrier}}(x)$ across this narrow junction.

**5.9.2 Wavefront Rectification** An incoming amplitude-modulated electromagnetic wave $F_{\text{wave}}(t)$ [5.5], propagating ballistically as a series of convective wavefronts [5.8], is intercepted by an external antenna coil. This induces a high-frequency alternating potential $V_{\text{RF}}(t)$ across the point-contact junction:

$$V_{\text{RF}}(t) = V_c\,[1 + m(t)]\cos(\omega_{\text{carrier}} t)$$

where: * $m(t)$ is the low-frequency audio modulation signal. * $\omega_{\text{carrier}}$ is the high-frequency carrier wave.

This potential shifts the internal frequencies of the mobile electron solitons ($\omega_{\text{metal}}$ in the graphite) relative to the localized valence states ($\omega_{\text{oxide}}$ in the iron oxide lattice) [5.6]:

$$\hbar\omega_{\text{metal}}(t) = \hbar\omega_{\text{oxide}} + eV_{\text{RF}}(t)$$

Due to the asymmetric profile of the Schottky barrier, the transition rate $\Gamma_{\text{trans}}$ of electron solitons [4.1] across the boundary is highly sensitive to the sign of $V_{\text{RF}}$:

1. **Forward Bias ($V_{\mathbf{RF}} > 0$):** The barrier width is compressed ($d_{\text{junction}} < \delta_{\text{contact}}$). The field envelopes of the electron loops in the graphite overlap with vacant states in the oxide semiconductor, resulting in a high transition rate and charge flow.
2. **Reverse Bias ($V_{\mathbf{RF}} < 0$):** The barrier width expands beyond the electron soliton's outer directrix radius ($d_{\text{junction}} \gg r_d$). This prevents field envelope overlap, shutting off the transition flow ($\Gamma_{\text{trans}} \approx 0$).

The asymmetric junction rectifies the current of electron solitons, governed by the non-linear diode relation:

$$I_{\text{rect}}(t) = I_0 \left( e^{\frac{eV_{\text{RF}}(t)}{k_B T}} - 1 \right)$$

This rectification converts the zero-mean alternating potential $V_{\text{RF}}(t)$ into a unidirectional, high-frequency pulsed current $I_{\text{rect}}(t)$ whose local average tracks the low-frequency modulation envelope.

**5.9.3 Low-Pass Filtration and Envelope Extraction** The rectified current $I_{\text{rect}}(t)$ is routed to a high-impedance telephone earphone. The receiving system forms a passive low-pass filter due to its inherent components: * The point-contact junction possesses a parasitic capacitance $C_{\text{junction}}$ in parallel with the semiconductor boundary. * The earphone consists of a coil with high inductance $L_{\text{phone}}$ and series resistance $R_{\text{phone}}$.

The high-frequency carrier component of the soliton current is bypassed through the low-impedance path of the junction capacitance $C_{\text{junction}}$.

However, because the audio modulation frequency is far lower than the carrier frequency ($\omega_{\text{audio}} \ll \omega_{\text{carrier}}$), the earphone coil presents low inductive reactance to the modulation envelope. The low-frequency current component $I_{\text{audio}}(t)$ flows directly through the earphone coil:

$$I_{\text{audio}}(t) \propto m(t)$$

This low-frequency current drives a varying magnetic field in the earphone coil, which exerts a fluctuating force on the earphone's metallic diaphragm. The diaphragm vibrates mechanically, releasing acoustic wavefronts that correspond to the original audio signal $m(t)$, completing the AM detection process.